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REVIEW

Aspirin responsive erythromelalgia in JAK2-thrombocythemia and incurable inherited erythrothermalgia in neuropathic Nav1.7 sodium channelopathy: from Mitchell 1878 to Michiels 2017

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ABSTRACT

Introduction: The article reports on the nosologic classification and common etiologic pathways of aspirin responsive erythromelalgia in clonal JAK2 thrombocythemia and incurable inherited autosomal dominant erythrothermalgia in Nav1.7 channelopathy.

Areas covered: Aspirin responsive platelet-mediated erythromelalgic arteriolar inflammation and thrombotic manifestations in thrombocythemia are caused by prostaglandin endoperoxides released from spontaneous activation, release reaction and aggregation of hypersensitive constitutively activated platelets due to gain-of-function mutations in the JAK2, TPO, MPL and CALR genes (sticky platelet syndrome). Incurable inherited erythrothermalgia is a rare autosomal dominant neuropathy due to gain function mutations in the SCN9A gene that result in warmth-induced burning pain of the body and both legs sparing the toes and fingers caused by hyperexcitable action potentials firing of the Nav1.7 mutated small C-fiber DRG nociceptive sensory neurons of the skin.

Expert opinion: Acetyl salicylic acid (aspirin) is a wonder drug in the treatment of symptomatic thrombocythemia patients with erythromelalgia but not sodium salicylate, coumadin and the platelet inhibiting agents ticlopidin and dipyridamole. The JAK2, MPL, CALR or TPO gain of function mutations constitutively activate megakaryopoiesis and increase the production of hypersensitive platelets as the cause of aspirin responsive erythromelalgia. Incurable inherited erythrothermalgia is a congenital autosomal dominant neuropathic Nav1.7 channelopathy of the afferent C-fibers neurons and blood flow regulating system in the arteriole venule shunt of the skin caused by a gain-of-function mutation in the SCN9A gene.

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Erythromelalgia; aspirin; primary erythromelalgia; inherited erythrothermalgia; congenital insensitivity for pain; thrombocythemia; platelet cyclo-oxygenase; SCN9A gene; Nav1.7 sodium channel; JAK2 gene

1. Introduction

Mitchell (1872) first reported on bilateral symmetric burning pain and red congestion in the feet in four young adults, aged 21–40 years, quite similar as described for primary erythromelalgia (PE) (Figure 1) [1]. The burning distress was elicited by exercise and warmth and relief was obtained by cooling of the affected extremities and rest. Mitchell (1878) reported 16 cases with painful affections of the feet of various types and severity in his article entitled *'On a rare vaso-motor neurosis of the extremities and on the maladies with which it may be confounded'*, like rheuma, lupus erythematoses, gout, Raynaud syndrome etc [2]. The erythromelalgic complaints begin with pain in feet with burning pain and red congestion symmetrically located in both feet and legs [1] or asymmetric limited region of one or both soles toes or fingers [2]. Mitchell labeled the painful foot and hand disorder erythromelalgia, deriving it from the Greek words *erythros* = red, *melos* = extremity and *algos* = pain [2]. A most typical description of erythromelalgia by Mitchell was a 58-year old man, who first presented with attacks of burning sensation and pain for 3 years. The burning pain in the right foot spread

from the big toe across the sole, beneath the metatarsophalangeal articulation to the right side and heel of the foot associated with a dusky mottled redness and painful on touch as described for erythromelalgia in PV patients (Figure 2) [2,3]. Mitchell and Spiller (1899) attributed erythromelalgia of both legs and feet to an undefined form of vasomotor neurosis or neuritis of the peripheral nerves or nerves within the spinal marrow [4]. Reviews on erythromelalgia by Cassirer (1912) [5], and May and Hillemand (1924) [6], put emphasis on the lack of characteristic clinical criteria to distinguish erythromelalgia from Raynaud syndrome, rheuma, gout, scleroderma and vasomotor disturbances. Osler in 1908 [7] and Brown in 1932 [8] described cases of secondary erythromelalgia in polycythemia vera (PV) in which pain in the hands and feet with the extreme red congestion was suggestive for erythromelalgia of Mitchell (1878) [2]. Brown studied 81 cases of which 12 with primary erythromelalgia, 10 with burning painful erythromelalgia secondary to polycythemia and 59 cases classified as atypical acroparesthesia without burning distress. Brown (1932) analyzed the 16 cases of Mitchell and his personal clinical observations, in which burning pain in the hands or feet was a prominent or disabling symptom.

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Article highlights

- Aspirin responsive erythromelalgia is caused by spontaneous activation and aggregation of hypersensitive JAK2-mutated platelets at high shear stress rate in the arteriolar endarterial circulation of the peripheral, cerebral, ocular and coronary circulation and can easily explain the occurrence of migraine-like cerebral ischemic attacks (MIAs) in clonal myeloproliferative thrombocythemia.
- Irreversible inhibition of platelet cyclooxygenase by aspirin (acetylsalicylic acid) completely relieved erythromelalgia for the duration of treatment, but sodium salicylate, dazoxiben, dipyridamol and ticlopidin did not inhibit platelet cyclo-oxygenase and had no effect at all on erythromelalgia indicating that activation of prostagandin metabolism by JAK2-mutated platelets at high shear stress in arterioles is necessary for erythromelalgia to develop.
- Erythromelalgia and its subsequent microvascular ischemic complications in JAK2, CALR, TPO and MPL mutated clonal myeloproliferative thrombocythemias is relieved and cured by aspirin due to irreversible inhibition of platelet cyclo-oxygenase activity but not by anticoagulation with coumarin, unfractionated heparin (UFH) or low molecular heparin (LMWH).
- Incurable erythrothermalgia is a rare dominantly inherited neuropathy caused by gain function mutations in the SCN9A gene that result in warmth-induced burning pain of the body and both legs sparing the toes and fingers caused by spontaneous induction of action potentials firing of the Nav1.7 mutated small C-fiber DRG nociceptive neurons of the neurosensory-vasomotor arteriolar-capillary-AVS functional unit in the skin.
- Inherited erythrothermalgia in Nav1.7 channelopathy induced burn pain, like fire in severe cases, and red congestion of the skin of legs and body with sparing of the toes and fingers. Neuropathic erythrothermalgia remains incurable and therefore recently labeled as the 'Man on Fire Syndrome'.

This box summarizes key points contained in the article.

Brown (1932) postulated five basic criteria for the diagnosis of primary and incurable variant of burning red congested extremities (Table 1) [8]. Attacks of bilateral or symmetric burning pain occurring in the leg feet and hands were unseparately linked with red flushing of the skin, initiated by standing or exposure to heat and relief was obtained by elevation and exposure to cold. During the attacks, the affected legs were flushed and congested, exhibited increased local heat and there was no treatment available. Smith and Allen (1938) substituted the term erythromelalgia for the descriptive term, erythermalgia, to denote the variability of disease manifestations and to emphasize the importance of heat = therme as the main eliciting factor and a most important feature of red, burning and painful extremities [12]. Erythromelalgia patients seen by Smith & Allen noted that one single dose of acetylsalicylic acid (aspirin 500 mg) produced marked relief of burning distress that persisted for a few days [12]. Since then both terms erythromelalgia and erythermalgia were used indiscriminately as synonyms for aspirin responsive or resistant primary and secondary forms.

The etiology of primary and secondary erythromelalgia/erythermalgia remained elusive until Michiels (1975–1985) discovered that aspirin responsive erythromelalgia in thrombocythemia (Table 2) was clearly distinct from the incurable primary erythromelalgia defined by Brown (Table 1) [3,11,19–22]. Michiels (1988–1996) described three distinct disease categories of erythromelalgia: aspirin responsive erythromelalgia in clonal thrombocythemia, incurable primary erythermalgia (PE), and aspirin resistant secondary erythermalgia [13–15,23].

Following the five postulates of Brown, Michiels defined the incurable and rare variant of red painful extremities in the complete absence of any underlying disorder as an incurable, congenital autosomal dominant disease entity and labeled it incurable primary erythermalgia: PE (Table 1, Figure 1) [13,14]. The clinical manifestations of incurable inherited PE were completely different from acquired aspirin-responsive erythromelalgia caused by platelet-mediated arteriolar inflammation and thrombosis in thrombocythemia of patients with essential thrombocythemia (ET) and PV (Table 2, Figure 2) [23]. Drenth & Michiels (1994) defined ten differential diagnostic criteria for incurable inherited PE to separate PE from aspirin responsive erythromelalgia in thrombocythemia (Tables 1 and 2) [16,24–28]. The burning pain, like fire in severe cases [29], of incurable inherited PE spontaneously arises in early or late childhood or adolescence with symmetric bilateral localization in legs and persists throughout life (Figure 1) [13,28]. There is relative sparing of the toes. Acrocyanotic ischemia or peripheral gangrene as the result of peripheral vessel (microvascular thrombotic) obstruction in skin biopsies in aspirin responsive erythromelalgia (Figure 2) [19] was never seen in skin punch biopsies from PE cases [29]. The histopathologic findings in skin biopsies from erythermalgic areas of PE patients were nonspecific showing the complete absence of any vascular or inflammatory underlying disorder and did not reveal a clue to its pathophysiology [29]. There are a multitude of clinical conditions without thrombocythemia associated with aspirin resistant bilateral and symmetric red swollen and burning pain in the feet and hands mimicking incurable inherited PE. Secondary variants of symmetric burning redness and warmth of the feet has been referred by Drenth & Michiels as secondary erythermalgia, which originate from side effects of drugs (e.g. verapamil) or may arise from disorders of undetermined significance in histopathology as documented in skin biopsies [29–33]. The literature on erythromelalgia and erythermalgia is very confusing since redness and pain of the feet are present in many other conditions. According to Mitchell (1878) and Michiels (2017) painful fingers, toes, hand and feet associated with distinct underlying diseases including Raynaud phenomenon followed by scleroderma or positive autoimmune serology, cutaneous vasculitis, lupus erythematoses, gout, atherosclerosis and diabetes may mimic but should not be labeled as secondary erythromelalgia or erythermalgia. Diabetic microvasculopathy during long-term follow-up may go through a painful erythermalgic-like stage of dysregulated vasomotor-neurosensory function of the endarterial small fiber neuron-arteriole-capillary-arteriole/venule/shunt (AVS) functional unit in the skin of feet and toes complicated by painful diabetic vasculopathy followed by insensitiveness to pain in the natural history of advanced or end-stage diabetic foot.

2. Aspirin responsive erythromelalgia in clonal thrombocythemia

2.1. Aspirin responsive erythromelalgia in thrombocythemia of ET and PV patients

Erythromelalgia in thrombocythemia is preceded by acroparesthesias felt as itching and prickling sensation often like pins

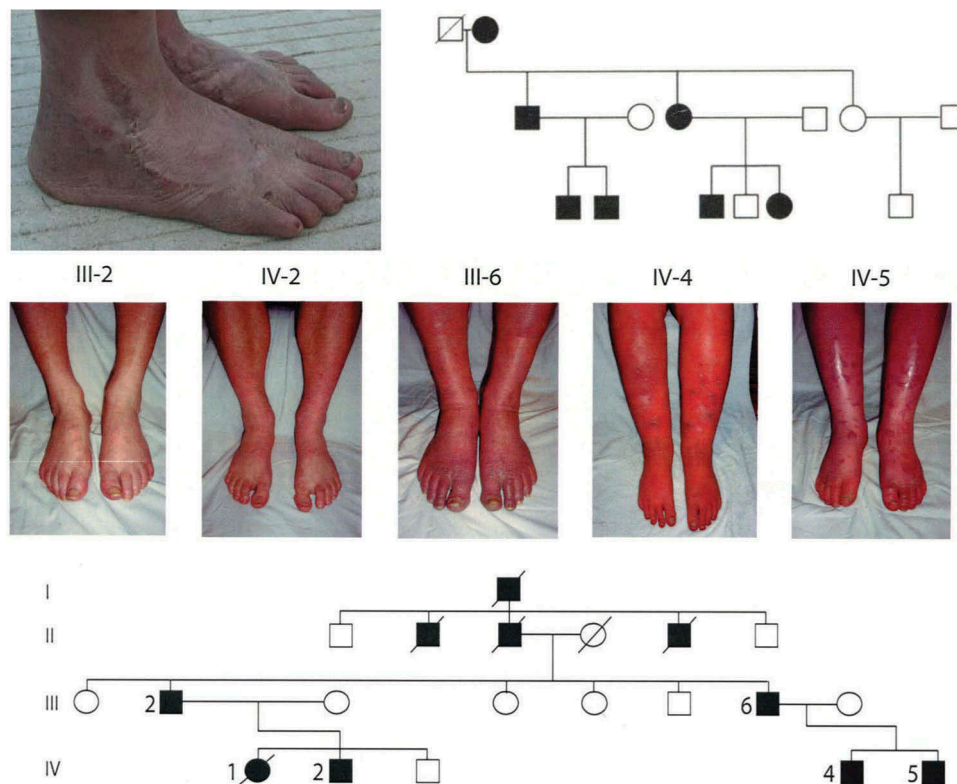


Figure 1. Clinical features of the proband in the family with typical primary erythromelgia (PE) showing diffuse symmetric redness of the feet and scars due to injury from repeatedly excessive cooling to relieve the severe erythromelgia like fire (cases IV-4 and IV-5) [9], in which the first autosomal dominant gain of function mutation T2573A in the SCN9A gene encoding for the L858 H mutation in the Nav1.7 sodium channel was detected in seven affected members in 3 generations of one PE family [10]. The L858 H mutant Nav1.7 voltage-gated sodium channel alpha subunit predominantly expressed in sensory and sympathetic dorsal root ganglia neurons (DRG), which are suspected to play an important role in nociception, vasomotor regulation and thermoregulation of blood flow in the skin (Figure 5). Autosomal dominant aspirin-resistant symmetric red warm and burning painful lower legs and feet in four generations of a Flemish family with incurable autosomal dominant primary erythromelgia (PE) diagnosed in 2005 as a neuropathic disorder of hyperexcitable small peripheral and sympathetic neurons caused by the gain of function SCN9A mutation S241 T in the voltage-gated sodium channel alpha subunit Nav 1.7 [9].

and needles, which in time are complicated by scorching and burning pain of erythromelgia associated with red congestion and bluish discoloration of the skin followed by digital ischemia or even gangrene (Figure 2). The asymmetric and unilateral localization of red congested, warm and painful swelling affects one or more toe, the sole of the forefoot or the tips of the fingers (Figure 2) [11,20]. The intensity of the pain correlates well with the degree of increased skin surface temperature. Burning pain arises when the thermographic skin surface temperature exceeded 31°C (Figure 2(b)) [34]. The relief of burning pain and red congestion for a few days by one loading dose of aspirin (500 mg) is a pathognomonic diagnostic clue of erythromelgia in thrombocythemia in its essential form of ET or associated with PV (Figure 3) [19–21]. The few days period of pain relief is similar to the few days duration that one dose of aspirin (500 mg) inhibits arachidonic induced platelet aggregation due to irreversible inhibition of platelet cyclooxygenase (COX-1) activity [20]. Reversible inhibition of platelet COX-1 activity by one dose of indomethacin 25 mg relieves erythromelgia in thrombocythemia for about 12 h. Coumadin and the platelet inhibiting agents ticlopidine, clopidogrel (platelet ADP-receptor inhibitor) dipyridamol and thromboxane synthetase inhibitors do not inhibit platelet COX-1 activity and are not effective in the treatment of aspirin responsive erythromelgia in thrombocythemia. Erythromelgia in thrombocythemia is caused by platelet-

mediated arteriolar inflammation and von Willebrand factor-platelet mediated ongoing inflammation and thrombosis [1,11,20]. In our understanding the very early initial stage of erythromelgia in clonal thrombocythemia induces a dysregulation of the vasomotor – neurosensory function unit of the endarterial small fiber neuron-arteriole-capillary-arteriole-venule shunt (AVS) in the skin of legs, feet and toes (Figure 4).

2.2. Complications of erythromelgia in thrombocythemia of ET and PV patients

If left untreated the initial red congested stage of erythromelgia in thrombocythemia of ET and PV patients progresses into blue ischemic spots in erythromelgic areas, acrocyanosis or even digital gangrene of one or more toes during long-term follow-up (Figure 2) [23,34]. In contrast to arteriosclerotic circulatory obstruction, peripheral arterial pulses usually remain normal in thrombocythemia patients with erythromelgia. The histopathological appearances of a skin punch biopsy from areas of recently relapsed erythromelgia 1 week after aspirin discontinuation show arteriolar lesions of fibromuscular intimal proliferation without involvement of venules, capillaries or nerves (Figure 3, left panels) [19–22]. The lining endothelial cells were swollen with large nuclei. The vessel wall was thickened by proliferation of cells, with narrowing of the lumen. Immunofluorescence studies with factor VIII antisera an affected

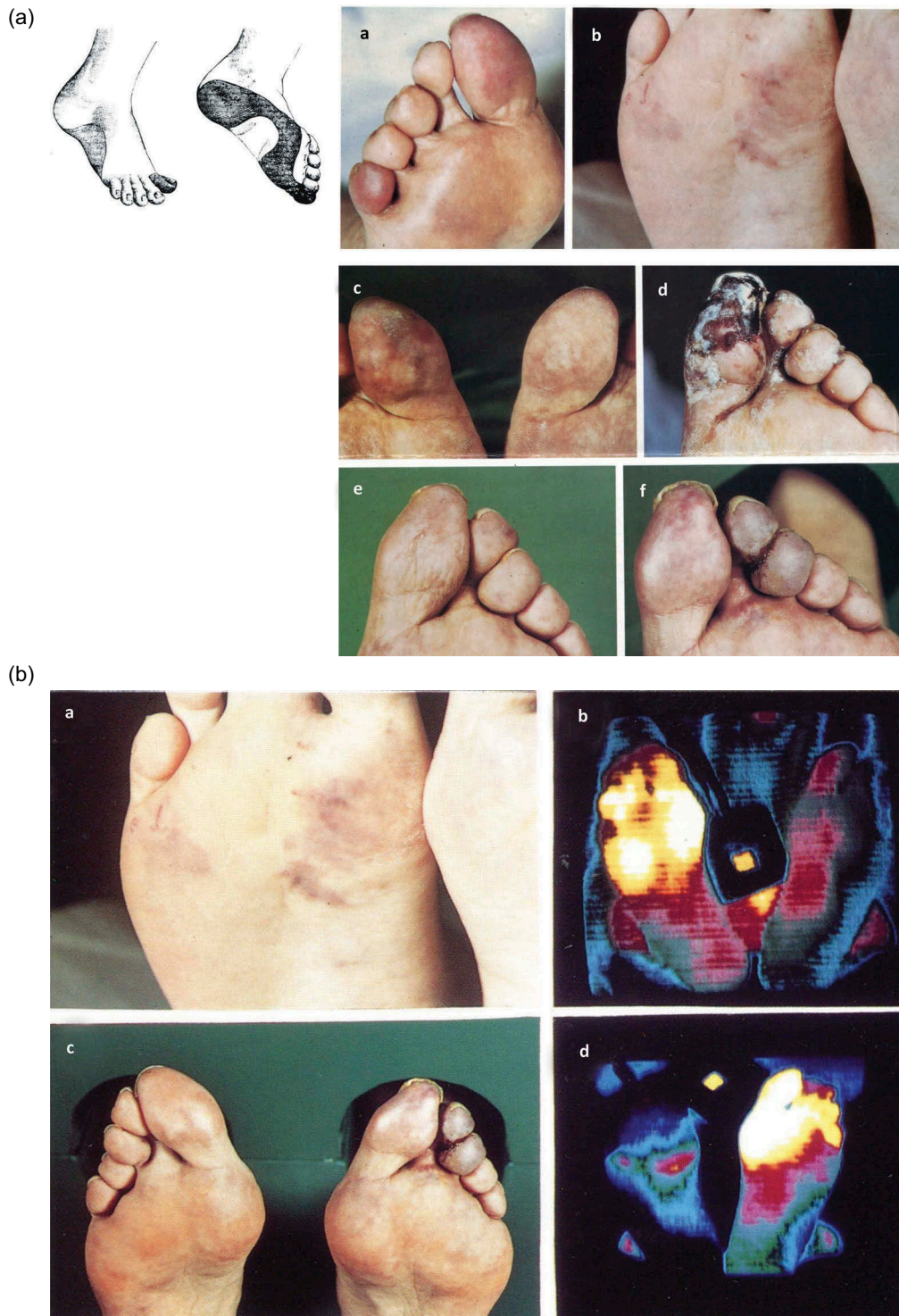


Figure 2. Black-white. Area of erythromelalgia in the right foot in case 3 J.P.S. described by Mitchell in 1878 in a 58-year-old man with a ten years history of asymmetric attacks burning pain, red congestion and bluish discoloration in the big toe, forefoot sole, lateral edge followed by several episodes of atypical transient ischemic attacks during four years follow-up. **Figure 2. (a)** Severe asymmetric burning pain and red warm congestion of big and little toe, and sole of forefoot of the right foot 1 week after discontinuation of aspirin. **B.** Burning, aching pain and mottled red-blue discoloration of the sole of the right foot 3 week after discontinuation of aspirin. **C.** Detail of an untreated painful and bluish big toe, of which the skin surface temperature is low (cold ischemic toe) [11]. **D.** Progression of painful toes to 'black' big toe and peeling of the skin of second and third toe in a 6 week period of coumarin treatment. **E.** Complete restoration of painful black ischemic toe within a few weeks by treatment with aspirin. **F.** Relapse of painful red and blue toes 2 weeks after discontinuation of aspirin [11]. **Figure 2. (b)** Typical mottled red blue congestion and thermographic visualisation of aspirin responsive erythromelalgia in the fore foot and toes in two cases of thrombocythemia A and B essential thrombocythemia; C and D polycythemia vera. Skin surface temperature: blue 24–25, green 26–27, purple 28–29, red 30–31, yellow 32 and white 33 °C.

Table 1. Twelve diagnostic criteria for incurable primary erythermalgia (PE) and inherited neuropathic erythrothermalgia (ETM) in Nav1.7 mutated sodium channelopathy.

1. Attacks bilateral symmetric burning pain spontaneously occurs in the feet or hands.
 2. The attacks are elicited by exposure to warmth or heat or initiated or aggravated by standing and exercise.
 3. Relief is obtained by exposure to cold and elevation of the affected extremities.
 4. During the attacks, the affected extremities are flushed and congested and exhibit increase local heat.
 5. The pathogenesis is unknown until 2004 and there is no treatment available until to date.
- Following the five postulates of Brown [8], the rare and incurable variant of red, painful extremities in the absence of any detectable underlying disorder has been recognized by Michiels as a distinct congenital clinical entity and therefore labeled as primary erythermalgia: PE [13–15]. In extension to the five postulates of Brown, Drenth & Michiels added four additional specific features of PE [16].
6. Primary erythermalgia (PE) spontaneously arises in children and adolescence as burning pain in both feet and ankles, and lower legs that persists throughout life.
 7. There is relative sparing of the toes and acrocyanotic ischemia or peripheral gangrene as the result of peripheral vessel obstruction is never seen. Trophic skin changes, ultimately ulceration, and even necrosis do not result from the disease, but are the consequence of excessive cooling.
 8. The inheritance of primary erythermalgia (PE) became clear by the demonstration of an autosomal dominant inheritance in several families and in one large family with 29 affected members in three generations. The early and spontaneous onset in life and the persistence throughout life support the congenital nature of primary erythermalgia.
 9. The histologic findings in skin biopsies from erythermalgic areas are nonspecific, showing the absence of an underlying disorder and thus do not reveal a clue to its pathophysiology.
 10. Drenth et al (2001) located the PE susceptibility gene on chromosome 2q:31–32 [17]. Yang et al. (2004) first discovered a heterozygous gain of function mutation in the SCN9A gene on chromosome 2q24- coding for the high voltage dependent sodium channel alpha subunit (Nav1.7) in seven affected family members in three generations with autosomal dominant inherited PE [10]. Nav1.7 incurable inherited primary erythermalgia (PE) has become the first dominant congenital painful neuropathic Nav1.7 sodium channelopathy.
 11. In this report the terms incurable congenital primary erythermalgia (PE) or inherited erythrothermalgia are replaced by the novel term of inherited neuropathic erythrothermalgia (ETM). Autosomal recessive homozygous or double null-missense ETM mutations in the SCN9A gene and Nav1.7 sodium channel membrane protein in DRG neurons is associated with congenital insensitivity to pain (CIP).
 12. The dominant inherited Nav1.7 channelopathies produce a continuum of two main clinical phenotypes of inherited erythrothermalgia (ETM) and paroxysmal erythromelalgic pain disorders (PEPD) and overlap ETM/PEPD disorders [18]. The broad spectrum of Nav1.7 channelopathies, ETM, PEPD and CIP will serve as a disease model for the development of rational, mechanism-based new treatment options for incurable pain.

Table 2. Twelve diagnostic criteria for aspirin responsive erythromelalgia in acquired JAK2 mutated thrombocythemia [11,19–22,34–45]: and in dominant inherited JAK2 mutated thrombocythemia [46–55].

1. Aspirin responsive erythromelalgia is characterized by red, warm congested extremities, and painful burning sensations, with preferential involvement of forefoot, sole and one or more toes and fingers.
2. Warmth intensifies the discomfort and cold provides relief.
3. Acroparesthesias (e.g. tingling, pins-and-needles sensations and numbness in toes and finger tips) usually precede the disabling burning distress of aspirin responsive erythromelalgia in clonal thrombocythemia.
4. The prompt relief of pain for a few days by a single dose of aspirin (500 mg) is so specific for aspirin responsive erythromelalgia in thrombocythemia that it can be used as a pathognomonic diagnostic criterion.
5. If left untreated, the initial red congested erythromelalgia usually progresses to intense burning throbbing, and aching with peeling of the skin of affected toes or fingers that subsequently become cold, ischemic or even gangrenous.
6. Aspirin responsive erythromelalgia is causally related to essential thrombocythemia and thrombocythemia associated with polycythemia vera (PV). Erythromelalgia is caused by platelet-mediated inflammation and thrombosis in the arteriolar endarterial circulation. Erythromelalgia in PV is not relieved by phlebotomy because the thrombocythemia persists.
7. The histopathologic findings in skin biopsies from affected skin are featured by fibromuscular intimal proliferation and von Willebrand factor-rich thrombotic occlusions (platelet thrombi) of the arterioles in the absence of preexistent vascular disease.
8. Complete relief of erythromelalgia and its ischemic circulatory disturbances is accomplished with low dose aspirin (50–100 mg/day) by its irreversible inhibition of platelet cyclooxygenase activity. Reducing the platelet count to normal level ($<350 \times 10^9/L$) abolishes the symptoms.
9. Shortened platelet survival in thrombocythemia complicated by erythromelalgia and its correction by curative treatment with aspirin indicates that the symptoms are provoked by intravascular platelet activation and aggravation *in vivo*, which preferentially takes place at high shear rate conditions in arterioles.
10. As aspirin responsive erythromelalgia does not occur in reactive thrombocytoses, it is postulated that not only increased platelet numbers above $400 \times 10^9/L$, but also an unknown qualitative disorder of platelet function (hyperreactive 'sticky platelets') in thrombocythemia of various myeloproliferative disorders leads to an increased risk of microvascular thrombotic complications not only involving the peripheral, but also the coronary and cerebral endarterial circulation (aspirin responsive platelet thrombophilia). Ref [20]
11. Hypersensitive sticky platelets in acquired JAK2 mutated thrombocythemia are the cause of aspirin responsive erythromelalgic circulation disturbances in ET and PV patients and have been observed in association with acquired mutated MPL and CALR thrombocythemias.
12. Hypersensitive platelet in dominant inherited thrombocythemias are the cause of aspirin responsive erythromelalgic circulation disturbances has been observed in relation with germ-line gain of function mutations in the thrombopoietin (TPO), JAK2, and MPL genes.

arteriole showed one inner single layer of fluorescent endothelial cells and with an antiserum against smooth muscle cells the thickened multilayer of cells in the media was strongly positive and diagnosed as platelet induced 'fibromuscular intimal proliferation' [19,20,22]. Fibromuscular intimal proliferation is not yet complicated by occlusive arteriolar thrombi in the early inflammatory red congested stage of erythromelalgia 1 week after discontinuation of aspirin (Figure 3, left panels).

Histopathology from a skin area of untreated relapsed erythromelalgia complicated by blue ischemic spots (Figure 2) after 3 weeks of aspirin discontinuation showed arterioles with pronounced 'fibromuscular intimal proliferation' complicated by

partial or complete occlusion of the lumen by organized thrombus in skin areas from bluish discolored erythromelalgic area (Figure 3, middle panels) [19,22,35]. The occlusive thrombi stained strongly positive for von Willebrand factor (VWF) opposed to no or weak fibrin staining (Figure 3, right panels), which is indicative for platelet thrombi formation [36].

2.3. Erythromelalgia caused by platelet-mediated arteriolar thrombosis in thrombocythemia

Platelet kinetic studies in 10 individuals with reactive thrombocytosis, in 10 patients with thrombocythemia complicated by

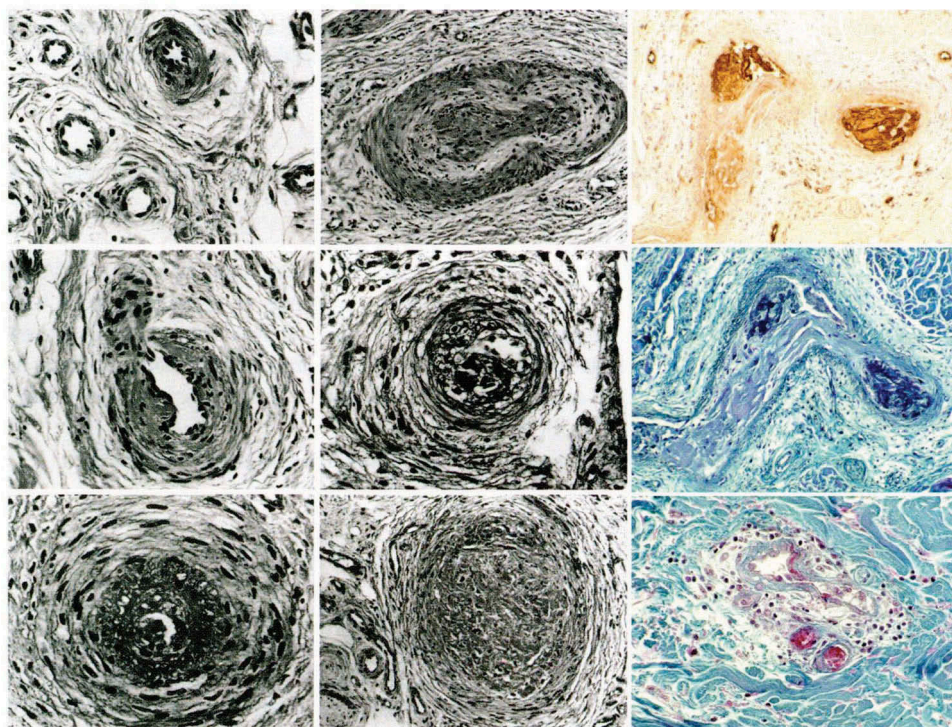


Figure 3. Skin punch biopsies from skin areas of recently relapsed erythromelalgia 1 week after discontinuation of aspirin (Figure 2(a)) in ET patients show arterioles with swollen endothelial cells and fibromuscular intimal proliferation (left) [19,20]. Skin punch biopsies from blue spots of skin areas with relapsed erythromelalgia 3 weeks after discontinuation of aspirin (Figure 2(b)) show arteriole with fibromuscular intimal proliferation with partial and complete occlusion of arteriole by fresh organized thrombus (middle upper two). Arteriole with complete occlusion of the arteriole inskin area of untreated erythromelalgia of an acrocyanotic melalgic toe (middle bottom) [19,20]. An arteriole occluded by a thrombus show a strong staining for von Willebrand factor (VWF, right upper) and a very weak staining for fibrin (right middle) indicating a platelet-rich thrombus. Positive fibrin staining of an arterial thrombus in a patients with lupus anticoagulant (right lower) [36].

erythromelalgia (E+) and in 10 patients with asymptomatic thrombocythemia (E-) [35,36] showed that platelet survival times of autologous Cr-labeled platelets in the E+ group (4.2 days) were significantly shorter than in the group with RT (7.7 days) and E-group (7.2 days). The platelet survival time in symptomatic E+ patients of 4.0 days significantly increased to 6.9 days during maintained aspirin demonstrating that erythromelalgia indeed is a platelet-mediated microvascular thrombosis [35]. Two symptomatic E+ patients on adequate oral anticoagulation therapy showed shortened platelet survival times, which reversed to normal by additional aspirin treatment of erythromelalgia [35]. The levels of endothelial cell marker thrombomodulin (TM), platelet activation markers platelet factor 4 (PF-4), β -thromboglobulin (β TG) were significantly increased as compared to asymptomatic ET (ET-). The five ET patients with erythromelalgia (E+) before and after aspirin treatment had significant higher β TG and TM levels before aspirin, but no significant differences were detected in either F1 + 2 or TDP levels before and after aspirin. The association erythromelalgia and significant increase of β TG and thromboxane B2 urinary (TXB2) excretion was tested in six ET patients after discontinuation of aspirin. Within 10 days after withdrawal of aspirin the development of endarterial (microvascular) VWF rich platelet thrombi in erythromelalgic areas was associated with a fourfold increase of β TG and a 3–30-fold increase in urinary TXB2 excretion [37,38]. Both the increased β TG and urinary TXB2 excretion and clinical signs of erythromelalgic microvascular thrombosis could be corrected with a platelet-specific very low dose aspirin regimen of 50 mg daily, which is sufficient

enough to inhibit platelet cyclooxygenase (COX-1) for the prevention of platelet thrombi in symptomatic thrombocythemia patients [37,38]. These real field observations made by Dr Van Genderen explain the rationale for the use of low maintenance dose aspirin 50–100 mg/day in clonal thrombocythemia and demonstrated the ineffectiveness of any other platelet inhibiting agents and coumadin derivatives in the prevention of VWF-platelet mediated erythromelalgia and acrocyanotic ischemic complications in ET and PV patients [35–37].

2.4. Aspirin responsive erythromelalgia and MIAs in clonal thrombocythemia

Thrombocythemia related clinical manifestations not only include aspirin sensitive erythromelalgia but also migraine-like atypical transient ischemic attacks (MIAs) as the main presenting symptoms in ET and PV patients [11,20,34,39–43]. The time lapse between the onset of MIAs and diagnosis of thrombocythemia ranged from 4 to 12 years simple because the MIAs are pathognomonic manifestations of thrombocythemia in ET and PV patients and usually overlooked by neurologists, internists and hematologists. In the 1990s we studied the nature of aspirin responsive neurological and visual microvascular MIAs in 56 ET patients [39,40]. Brain CT, carotid duplex ultrasonography and extensive cardiac investigations at time of MIA or visual disturbances did not reveal any coexistent vascular cause, the main reason why such complaints are labeled as specific functional vasomotor disturbances of unknown etiology. MIAs are featured

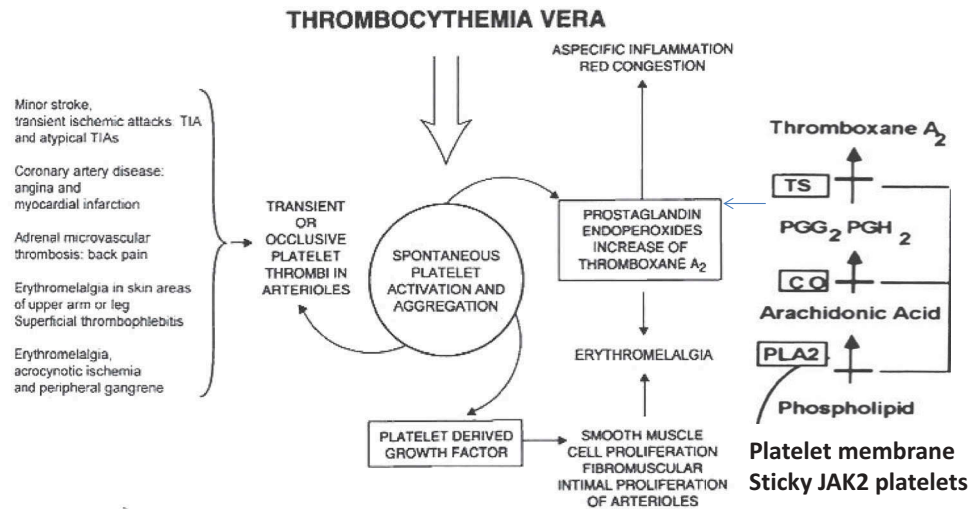


Figure 4. The broad spectrum of arteriolar inflammatory, fibromuscular intimal proliferation and occlusive platelet thrombi as the underlying histopathology of erythromelgia in JAK2 thrombocythemia vera and atypical migraine-like cerebral and ocular attacks (MIAs), acute coronary syndrome, adrenal microvascular thrombosis and superficial thrombophlebitis (erythromelgia of the skin in upper legs) are caused by spontaneous activation of sticky platelets at high shear in the endarterial peripheral, cerebral, and coronary circulation, which all together reflect a novel distinct disease entity of aspirin-responsive arterial thrombophilia in thrombocythemia vera in JAK2-mutated myeloproliferative neoplasms [3,19,20,23,34–43].

by poorly localizing, with unsteadiness, dysarthria and bilateral blurring of vision being the most common and focal symptoms were found in half of them. The focal symptoms, such as transient mono- or hemiparesis or monocular amaurosis fugax occurred sequentially, not simultaneously. The recurrent episodes all had a sudden onset, lasted for a few seconds to several minutes and were usually associated with a dull pulsatile headache. This clinical presentation of microvascular disturbances is atypical for transient ischemic attacks caused by atherosclerosis, but the striking similarity to migraine accompaniments supports the critical role of platelets in the pathogenesis of aspirin responsive headache and visual or cerebral ischemic symptoms in patients with ET and PV. If left untreated, the migraine-like atypical TIAs (MIAs) do progress to minor or major stroke featured by neurological deficits and a positive MRI scan of the brain [39,42]. Major vascular complications frequently occur in thrombocythemia associated with PV due to increase of leukocytes, hemoglobin hematocrit and erythrocyte counts above $6 \times 10^9/\text{L}$ [34,56]. Venesection aiming at a hematocrit around 0.40 in PV patients significantly reduced the high risk of major arterial thrombotic complication [34,41–58]. The erythromelalgic microvascular ischemic syndrome and MIAs persist if the PV is brought into hematological remission by phlebotomy alone when not on aspirin [57], because JAK2 mutated thrombocythemia persist with platelet count above $350 \times 10^9/\text{L}$ [41,44,45].

2.5. Platelet induced arteriolar inflammation in aspirin responsive erythromelgia

The immediate relief and cure of full blown inflammatory signs of erythromelgia by aspirin is due to irreversible inhibition of platelet cyclooxygenase activity (Figures 2). The inflammatory signs of burning pain appears when the skin temperature exceeds 31°C [11,20,21,40,42]. Superficial thrombophlebitis and livedo reticularis are skin manifestation of aspirin responsive erythromelgia in ET and PV patients [3].

The incidence of aspirin responsive superficial thrombophlebitis was 14% in our cohort of 50 thrombocythemia (30 ET and 20 PV) patients [40]. Michiels et al. measured the prostaglandin activity in artificial skin blisters in two ET patients from in skin area after effective treatment with aspirin and in control (Table 3) [59]. The total prostaglandin activity in fluid of skin blisters from thrombophlebitic erythromelalgic skin areas was very high as compared to control (Table 3) [59]. Effective treatment of erythromelalgic skin areas with aspirin was associated with reduction of total prostaglandin content in skin blisters to normal (Table 3). In subsequent experiments, we measured the effect of thromboxane A₂ inhibitor dazoxiben on the content of prostaglandin-E-like activity in artificial blisters from erythromelalgic skin areas in a case of thrombocythemia complicated by erythromelgia. Dazoxiben did not alleviate the erythromelgia in thrombocythemia and did not inhibit prostaglandin content in artificial blisters despite complete inhibition of platelet malondialdehyde and thromboxane B₂ synthesis (Table 4). During a 10-day dazoxiben treatment period, persistent erythromelgia was associated with a significant shortened mean platelet survival (3.2 days) [59]. Subsequent treatment with low dose aspirin daily completely relieved the erythromelgia, inhibited the prostaglandin endoperoxide production and corrected mean platelet survival to normal (7.9 days). We could conclude that increased prostaglandin endoperoxide activity in fluid from artificially skin blisters from erythromelalgic skin can be regarded as the cause of the inflammatory painful component of platelet-mediated erythromelgia (Figure 4) [59]. The analgesic effect of acetylsalicylic acid (aspirin) on the burning pain and vasomotor disturbances also lasted for 2–3 days and runs parallel with the irreversible inhibition of platelet cyclooxygenase (COX-1) [38]. These observations strongly indicate that release of prostaglandin endoperoxide metabolites, serotonin, platelet derived growth factor (PDGF) during platelet activation and aggregation at high shear rate in the endarterial

circulation are responsible for the inflammatory red congestion and fibromuscular intimal proliferation of early stage erythromelalgia (Figure 4). A number of inflammatory mediators, such as prostaglandin [60], adenosine [61] and serotonin [62] released by activated platelets in arterioles in patients with erythromelalgic thrombocythemia affect the electrophysiological properties of voltage-gated sodium channel Nav1.7 protein expressed in microvascular myocytes, endothelium and innervating dorsal root ganglion neurons of the arteriole-capillary-arteriole-venule-shunt vasomotor – neurosensory unit in the human skin (Figure 5) [63,64]. The von Willebrand factor (VWF) is highly expressed in endothelial cells of small arteries and arterioles as compared to endothelial cells of large arteries and veins for functional biological reasons to interact with platelets when activated at high shear stress conditions in the endarterial microcirculation. All the inflammatory mediators released during platelet activation and reversible aggregation in the arteriolar circulation are predicted to activate the Nav1.7 channels nociceptors of the afferent small fibers of the neurovasomotor, arteriole-capillary-arteriole-venous shunt thermoregulatory control system of skin blood flow (Figure 4). Platelet activation and clumps (reversible aggregates) at high shear rate in the endarterial circulation followed by release of prostaglandin endoperoxides, serotonin, adenosine that can sensitize nociceptive neurons are the explanation

Table 3. Prostaglandin E-like (PGE) activity in fluid of artificial blisters from skin areas with erythromelalgia (E+) in clonal thrombocythemia and control without erythromelalgia (C→E-) and from skin areas after treatment of erythromelalgia with aspirin (E+→E-) [59].

Blister fluid	Total PGE	PGE/mL
1.19 mL E+	6.43 µg	5.40 µg
1.29 mL E+	8.42 µg	6.53 µg
1.02 mL C→E-	0.82 µg	0.80 µg
1.36 mL E+→E-	1.35 µg	0.99 µg

Table 4. The effect of dazoxiben (400 mg every 6 hours for 10 days) on platelet malondialdehyde (MDA, nmol per 10⁹/L platelet: nmol/p) synthesis, plasma thromboxane B2 (TXB2) and prostaglandin E2 (PGE2) levels in a patient with essential thrombocythemia who suffered from erythromelalgia during the 10 days period of investigation [59]. Dazoxiben largely inhibited MDA production and TXB2 synthesis but did not inhibit PGE2, which was associated with the persistence of erythromelalgia.

Day	MDA nmol/p	TXB2 pg/mL	PGE2 pg/mL
	<i>p</i> *	<i>p</i>	<i>p</i>
	2 h**	2 h	2 h
1	20.1	1667	95
3	1.7	<15	98
9	0.2	<15	85

*pretreatment value.
**time lapse in hours after intake of dazoxiben.

of burning pain and red congested swelling in areas of aspirin responsive erythromelalgia and the migraine component of

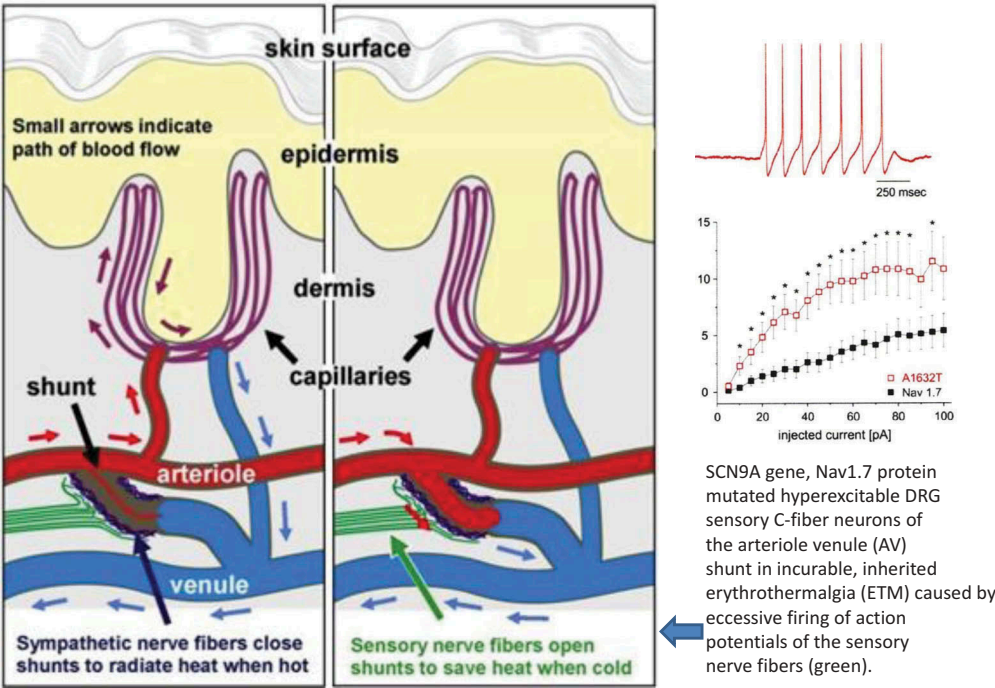


Figure 5. The arteriole capillaries venule functional unit of the skin related to **Erythromelalgia** caused by platelet-mediated shear stress induced arteriolar inflammation and thrombosis in thrombocythemia and to **Erythrothermalgia** caused by a gain of Nav1.7 function mutation in the C-fiber sensory nerve of the arteriole venule shunt inducing vasodilatation and hot burning pain: The Man in Fire Syndrome. Aspirin responsive erythromelalgia is caused by platelet cyclo-oxygenase mediated inflammation and red congestion and fibromuscular intimal proliferation, which takes place at high shear rate in the arterioles but the capillaries, arteriole venule (AV) shunt and nerves are not involved (Figures 3 and 4). Platelet prostaglandin endoperoxides synthesized by cyclo-oxygenase during platelet activation account for the inflammatory symptoms of red congestion and the secretion of platelet derived growth factor (PDGF) during platelet aggregation account for the fibromuscular intimal proliferation followed by the formation of occlusive platelet thrombi in arterioles if aspirin-responsive erythromelalgia is left untreated. Gain of function mutations in the SCN9A gene are the cause of Nav1.7 channelopathy in all cases with incurable inherited erythrothermalgia (ETM) and/or paroxysmal extreme pain disorder (PEPD (Table 5). In all described ETM and PEPD cases, skin punch biopsies do not shown morphological abnormalities of the arteriole, capillaries and venules [13,14,29]. The sensory DRG neuron afferent C-fibers (sensory nerve fibers in green) of the AV shunt in ETM and PEPD patients display hyperexcitability DRG neurons (afferent C-fibers) showing spontaneous firing of action potentials as the cause of incurable inherited erythrothermalgia and PEPD (Table 5), the Man in Fire Syndrome.

Table 5. Thirty-four SCN9A gain of function mutations and comparative analysis of NAV1.7 electrophysiology of hyper-polarized activation, slow inactivation, fast inactivation and ramp current of Nav1.7 mutant transfected HEK cells and resurgent currents and firing action potentials in Nav1.7 mutant transfected DRG in family members with early and late onset incurable primary erythralgia (PE), inherited erythrothermalgia (ETM), or paroxysmal extreme pain disease (PEPD).

Mutation	Domain	Activation ΔV_{half} (mV)	Slow inact	Fast inact/ Res Curr DRG	Ramp currents	DRG firing	Age at onset (years)	PE/IEM vs PEPD	Reference
Q10 R	D1S1	5.3 -	4.8 -	Ns/nd	Yes	Pos	14	PE/ETM	[65]
Q10 K	D1S1	Nd	Nd	Nd	Nd	.	0–20	PE/ETM	[66]
I136 V	D1S1	5.7-	8 -	Ns/nd	Yes	Nd	9, 22	PE/ETM	[67–69]
R185 H	D1S1S2 L	.	.	Res Curr	No	Pos	23, 24	PE/PEPD	[70,71]
S211P	D1S4	8.2-	15 -	Ns/nd	Yes	Nd	15	PE/ETM	[72]
F216S	D1S4	11.8 -	15.6 -	Ns/nd	Yes	Pos	36	PE/ETM	[73,74]
I228 M	D1S4	Ns	6.8 +	Ns	NT	Pos	32	PE/PEPD	[18]
I234 T	D1S4S5 L	18 -	18 -	Ns/nd	Yes	Nd	<1	PE/ETM	[75,76]
S241 T	D1S4S5 L	8.4 -	12.3 -	Ns/nd	Yes	Pos	2, 10	PE/ETM	[9,77]
L245 V	D1S4S5 L	Ns	Incompl	Abn	Yes	Nd	Child	PE/ETM	[78]
N395 K	D1S6	7.7 -	7.7 -	Ns/nd	NT	Nd	<10	PE/ETM	[79,80]
V400 M	D1S6	6.5 -	6.5 -	7.3 +/-nd	Yes	Nd	<1	PE/ETM	[81,82]
G616 R	D1D2 L1	Ns	Ns	5.7+/-nd	Yes	Pos	6, 8, 20	PE/ETM	[83]
D623 N	D1D2 L1	Ns	Impaired	Impaired	No	Pos	22	PE/ETM	[70]
I720 K	D1D2 L1	Ns	Impaired	Ns	No	Pos	37	PE/ETM	[70]
I739 V	D2S1	Ns	Impaired	Ns	No	Pos	14	PE/ETM	[70,84]
L823 R	D2S4	14.6 -	14.6 -	9.8 -/nd	NT	Nd	<1	PE/ETM	[85]
I848 T	D2S4S5 L	13.8 -	13.8 -	Ns/nd	Yes	Pos	4, 8	PE/ETM	[69,79,86]
G856D	D2S4S5 L	9.3 -	9.3 -	Res Curr	Yes	Pos	<10	PE/PEPD	[71,87]
L858 H	D2S4S5 L	13.3 -	13.3 -	Ns/nd	Yes	Pos	4, 8	PE/ETM	[10,86,88]
L858 F	D2S4S5 L	9 -	9 -	3.1 +/-nd	Yes	Pos	0, 2	PE/ETM	[79,89]
A863P	D2S5	8 -	8 -	10 +/-nd	Yes	Pos	Child	PE/ETM	[90]
V872G	D2S5	9.3 -	9.3 -	Ns/nd	Yes	Nd	5	PE/ETM	[91]
Q875E	D2S6	17.5 -	17.5 -	Ns/nd	Nd	Nd	2	PE/ETM	[92,93]
Del-L955	D2S5	24 -	24 -	Ns/nd	Yes	Pos	15	PE/ETM	[94,95]
M932 L/V991 L	D2S6	Ns	Ns	Res Curr	No	Pos	16	PE PEPD	[70,71]
R1279P	D3S4	18.1 -	18.1 -	9.6 +	Yes	Pos	18, 33	PE/ETM	[96]
V1298 F		Ns		Res Curr	Minor	Pos		PEPD	[97]
P1308 L	D3S4S5 L	9.6 -	9.6 -	16.1 +/-nd	Yes	Pos	2	PE/ETM	[97]
V1316A	D3S5	9.3 -	9.3 -	Ns/nd	Yes	Nd	9	PE/ETM	[98]
F1449 V	D3S6	7.6 -	7.6 -	4.3 +/-nd	No	Pos	3	PE/ETM	[99]
M1532I	D3D4 L3	.	.	/nd	No	Pos	68	PE PEPD	[70]
W1538 R	D4S2	8.6 -	.	Ns	Nd	.	3		[100]
A1632E	D4S5	Ns	7 -	Res Curr	Yes	Pos	birth, 3	PEPD ETM	[101]
A1632 T	D4S5	.	.	No Res Curr	No	Pos	3, 17	PE/ETM	[101]
A1632E	D4S5	.	.	/nd	Yes	Pos.	Birth→3	PEPD ETM	[70,101,102]
A1746G	D4S6			/nd	Nd		61	PEPD	

Ns: not significant; Nd: not done; TS: thromboxane synthetase; PGG2PGH2: prostaglandine endoperoxides G2 and H2; CC: cyclooxygenase synthetase; PLA2: phospholipase A2.

MIAs in JAK2^{V617F} mutated thrombocythemia (Figures 4 and 5). This pathobiological concept of aspirin responsive erythromelalgia is completely in line with increased plasma levels of platelet activation markers PF4, btg, TM and PDGF in thrombocythemia complicated by erythromelalgia, microvascular thrombosis and MIAs [19,39,40,42]. Increase of PDGF released from platelets and the endothelial cell (EC) activation marker TM [36,37] refers to increased platelet-EC-vessel wall interactions underlying the etiology of fibromuscular intimal proliferation (Tables 2 and 3, Figures 2 and 3) [19,20,22]. The platelet inhibiting drugs, sulfinpyrazon, dipyridamol, ticlopedine, clopidogrel and Coumadin do not inhibit platelet COX-1 activity and had no effect on erythromelalgia or MIAs [20,42].

3. Molecular pathobiology of erythromelalgia in JAK2 mutated thrombocythemia

3.1. Molecular pathobiology of erythromelalgia and MIAs in JAK2-positive ET and PV patients

Erythromelalgia in JAK2-thrombocythemia is provoked by spontaneous activation and aggregation of hypersensitive (sticky) platelets at high shear stress rate in the endarterial circulation of the peripheral, cerebral, ocular and coronary circulation and can easily explain the occurrence of migraine-like cerebral ischemic attacks (MIAs) in thrombocythemia [1–21,35–42]. Aspirin responsive erythromelalgia MIAs are never observed in reactive thrombocytosis. This clearly indicates that not only a quantitative (platelets

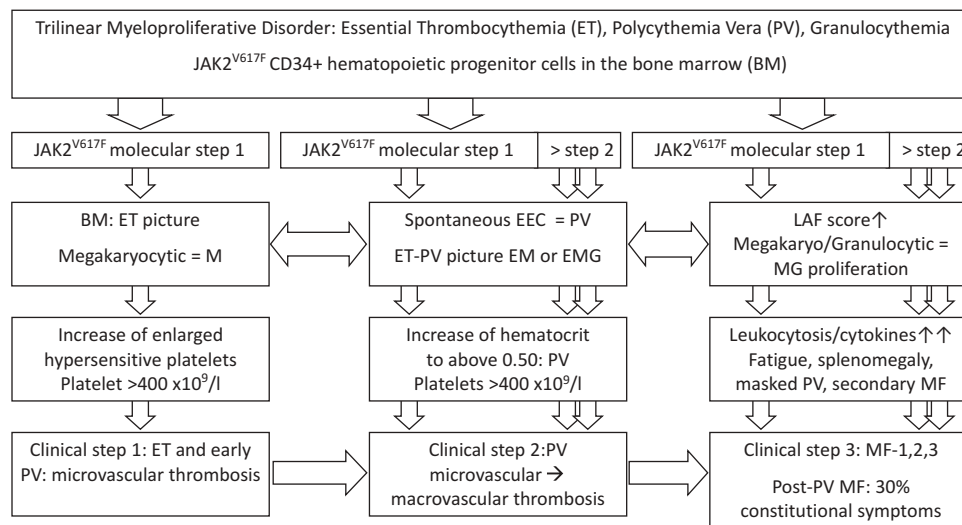


Figure 6. The molecular etiology of platelet mediated microvascular and major thrombosis, increased red cell mass, splenomegaly, and secondary myelofibrosis in JAK2^{V617F} mutated myeloproliferative neoplasm (MPN). The JAK2^{V617F} gain of function mutation in trilinear myeloproliferative neoplasms (MPN) is detectable in platelets, granulocytes and bone marrow hematopoietic stem cells and all stages of erythrocytic, megakaryocytic, and granulocytic progenitor cells. Progression of heterozygous (Step 1) of benign JAK2 essential thrombocythemia (ET) into heterozygous/homozygous or homozygous (Step 2) mutated stages of polycythemia vera (PV) due to mitotic recombination of chromosome 9p profoundly change the clinical, molecular and pathological (CMP) biological phenotype of ET into sequential stages of classical PV and advanced PV complicated by splenomegaly and secondary myelofibrosis (MF) [103,104]. The discovery of the somatic JAK2^{V617F} mutation as the driver cause of trilinear myeloproliferative disorders (MPNs) can explain that the gradual progression of JAK/Stat kinase activity at the molecular level of heterozygous into homozygous JAK2^{V617F} mutated MPN is associated with the sequential occurrence of ET, PV and MF phenotypes [105] and its extension to the clinical level with therapeutic implication for the use of low dose aspirin [44]. Aspirin responsive erythromelalgia and migraine-like ocular and cerebral ischemic attacks (MIAs) are the presenting symptoms of JAK2^{V617F} thrombocythemia caused by spontaneous activation and aggregation of hypersensitive JAK2-mutated platelets at high shear stress rate in the end-arterial circulation of the peripheral, cerebral, ocular and coronary circulation. Irreversible inhibition of platelet cyclooxygenase by aspirin (acetylsalicylic acid) completely relieved erythromelalgia and MIAs for the duration of treatment, but sodium salicylate, dazoxiben, dipyridamol, ticlopedin, coumarin and heparin had no effect at all indicating that activation of prostaglandin metabolism by JAK2-mutated platelets at high shear stress in arterioles is necessary for erythromelalgia and MIAs to develop. If left untreated erythromelalgia and MIAs are at high risk for major arterial thrombotic complications of digital acrocyanosis, TIA, stroke, and acute coronary syndrome. EEC: endogenous erythroid colonies; Erythrocytic megakaryocytic (EM); Erythrocytic megakaryocytic granulocytic (EMG) or megakaryocytic granulocytic (MG) proliferation; MF: myelofibrosis.

>350 × 10⁹/L), but also a qualitative 'hypersensitive' functional platelet defect (JAK2-mutated 'sticky' platelets) must be the underlying etiology of platelet mediated arterial thrombophilia in myeloproliferative clonal thrombocythemia [40–42]. A gain of platelet function from normal to hyperreactive or 'sticky' platelets in clonal thrombocythemia is postulated by Michiels & Van Vliet (1975–1996) as the underlying pathophysiology of platelet-mediated arteriolar inflammatory manifestations in the extremities and of MIAs [19–22,35–42]. With the advent of the JAK2^{V617F} mutation as the driver cause of clonal thrombocythemia in patients with ET and PV [103,104], it became evident that JAK2^{V617F} mutated megakaryocytes and platelets are constitutively activated and hypersensitive to thrombopoietin (TPO), and consequently do produce 'sticky platelets' as the cause of platelet mediated arteriolar inflammation (platelet thrombophilia [42]) in thrombocythemia (Table 5 and Figure 2) [105–107]. Heterozygous JAK2^{V617F} mutation is associated with no or slight increase of erythropoiesis in bone marrow biopsies of ET patients (Table 5) [45,103,104]. Homozygous JAK2^{V617F} mutation leads to trilinear erythrocytic, megakaryocytic and granulocytic (EMG) myeloproliferation [108] due to hypersensitivity of homozygous mutated hematopoietic progenitor cells to EPO and G-CSF (Figure 6) [103–105]. The sequential occurrence of erythromelalgic microvascular ischemic symptoms in heterozygous JAK2^{V617F} mutated ET and major arterial and venous thrombosis in homozygous JAK2^{V617F} mutated PV is related to increase of the JAK2 mutation load (Figure 6) [44,45]. About

30–40% of JAK2 mutated myeloproliferative neoplasms present with masked PV with significant splenomegaly and will develop post-PV myelofibrosis [44,45,108]. The association of erythromelalgia and MIAs has also been observed in our patients with congenital TPO, JAK2 and MPL hereditary essential thrombocythemia (HET) and the acquired somatic MPL, CALR and heterozygous JAK2^{V617F} mutated thrombocythemias [106–108]. These gain of function mutations constitutively activate megakaryopoiesis via the TPO-receptor (= MPL-receptor) signaling mechanism to an increased production of hypersensitive platelets as the cause of aspirin responsive platelet mediated arteriolar inflammation and thrombosis [108]. Erythromelalgia and acrocyanotic ischemic digital complications in JAK2-thrombocythemia (erythromelalgic thrombotic thrombocythemia: ETT) occur already at platelet counts above 400 × 10⁹/L [11,20,34–44,108]. Unfortunately, the early functional 'vasomotor' disturbances of erythromelalgia and MIAs are unknown to and overlooked by internist, hematologists and neurologists as the only or main presenting symptom of clonal thrombocythemia and consequently not treated with aspirin until ET and PV patients had developed acrocyanosis, digital ischemia (black toe, Figure 2), transient ischemic attacks, minor or major stroke, myocardial infarction or acute coronary syndromes [42]. At increasing platelet count from below to above 1000 × 10⁹/L during follow-up ETT is complicated by spontaneous hemorrhages (hemorrhagic thrombocythemia: HT or paradoxical ETT/HT) due acquired von Willebrand syndrome type 2A indicating the compelling

need to reduce platelet count to below one million or normal preferentially by pegylated interferon (Pegasys[®]), or anagrelide in low risk ET and PV and by hydroxyurea in high risk ET and PV [11,34,44,108]. CALR and MPL-mutated thrombocythemia have no features of PV and usually present with high platelet counts around or above $1000 \times 10^9/L$ complicated by paradoxical ETT/HT or HT.

3.2. Erythromelalgia and MIAs in dominant hereditary essential thrombocythemia

The first well documented report on autosomal dominant congenital hereditary essential thrombocythemia (HET) due to a gain of function mutation in the TPO gene was first described in a large Dutch family [47,108] and confirmed in a Polish family [48]. Analysis of the TPO gene in these two TPO-mutated HET families by Dr. Skoda revealed the same C→G transversion in the splice donor of intron 3 of the TPO gene on chromosome 3q27 in the affected with dominant inheritance in the Dutch and Polish HET families [46–48]. This mutation destroys the splice donor site in intron 3 and results in exon 3 skipping. The shortened 5'UTR gene resulted in a gain of function which leads to overproduction of TPO by a mechanism of increased efficiency of the TPO mRNA translation. The *in vivo* increased TPO levels are responsible for the etiology of congenital ET by stimulating megakaryocyte and platelet production both *in vitro* and *in vivo*. Congenital gain of function mutation in the TPO gene in the Dutch and Polish families resulted in increased plasma TPO levels, which induce a physiological activation of the TPO→MPL signaling pathway with persistent increase of hypersensitive 'sticky platelets' associated with aspirin responsive erythromelalgic microvascular circulation disturbances and atypical transient ischemic attacks similar as in acquired thrombocythemia (Figure 4) [20,21,46–48]. Three germ-line heterozygous gain of function mutations in the JAK2 gene, JAK2^{V617I}, JAK2^{R564Q} and JAK2^{H698N} has been discovered as the cause of congenital dominant hereditary essential thrombocythemia (HET) [49–52]. The two germ-line mutation JAK2 mutations were the sole genetic abnormality, sufficient to induce the HET phenotype without features of PV complicated by erythromelalgia and microvascular ischemic events in affected members of congenital thrombocythemia [49,50].

4. Primary erythromelalgia or inherited erythrothermalgia, paroxysmal extreme pain disorders and congenital insensitivity to pain in neuropathic Nav1.7 channelopathy

4.1. Incurable inherited primary erythromelalgia (PE) in neuropathic nav1.7 channelopathy

Drenth et al. (1991) localized the incurable inherited PE susceptibility gene on the long-arm of chromosome 2 (2q34.3) [17] in a cohort of European [29] PE families and the large Finley PE family in the United States [28] with PE and proven autosomal dominant inheritance in two to four generations (Figure 1). Yang et al.

(2004) studied a Chinese family with PE in three generations and first discovered a heterozygous gain of function mutation (L858 H) in the SCN9A gene on chromosome 2q24.3 coding for the high voltage dependent sodium channel alpha subunit (Nav1.7), which segregated with the affected family members (Figure 1) [10]. Heterozygous gain of function mutations in the SCN9A gene as the cause of incurable inherited primary erythromelalgia has been immediately confirmed in five European PE families by Drenth et al. [79], in a Flemish PE family by Michiels et al. (Figure 1) [9], and in the large USA PE family in five generations by Waxman et al. [99]. Drenth et al. detected four gains of function SCN9A mutations (I848 T, N395 K F610 T, L858 F) in the five PE European families. Heterozygous gain of function mutation F1449 V in the SCN9A gene (Nav1.7) was found in 14 of 14 affected members of the large USA family. Michiels et al. detected a novel heterozygous S241 T mutation in the SCN9A gene (Nav1.7) in all five investigated affected members in three generations of a Flemish family (Figure 1) [9]. Drenth and Waxman (2007) highlighted the gain of function mutations in Nav1.7 encoded by the SCN9A gene that cause a spectrum of human pain disorders of incurable inherited primary erythromelalgia (PE) and paroxysmal extreme pain disorder (PEPD) [110]. Waxman and his team of PE investigators collected 29 gain of function mutations in the SCN9A gene coding for the Nav1.7 voltage sodium channel as the cause of incurable inherited PE (N = 19), paroxysmal extreme pain disorder (PEPD, N = 10) or small-fiber neuropathy (SFN, N = 7) (Figure 5) [110–112]. Tang et al. reviewed the causative gene 35 SCN9A mutations and the Nav1.7 sodium channels' electrophysiological properties in 26 cases of incurable inherited erythromelalgia [113]. All Nav1.7 gain of function mutations detected in PE and PEPD patients are missense mutations that change important and highly conserved amino acid residues of the Nav1.7 protein (Table 5) [84,110]. Incurable inherited PE became the first dominant congenital painful neuropathic Nav1.7 sodium channelopathy as a model disease for other painful conditions and for the development of rational, mechanism-based treatment options for pain [10–112].

Waxman used the term inherited erythromelalgia IEM) [111,112] for incurable inherited PE as defined by Drenth & Michiels in Table 1 [13,14,16,26,110]. Using the same term erythromelalgia for the nosologic classification of platelet mediated and neuropathic induced burning, painful red congested extremities is confounding and not scientific. Aspirin responsive erythromelalgia occur in all variants of TPO, MPL, JAK2, and CALR and heterozygous acquired and congenital JAK2 mutations as driver causes of thrombocythemias. All these gain of function mutations constitutively activate megakaryopoiesis in the bone marrow through activation of MPL-receptor (TPO-receptor) signaling pathway with increase of hypersensitive platelets as the cause of platelet-mediated erythromelalgia, ETT and HT. The inflammatory and microvascular thrombosis of erythromelalgia in acquired and congenital clonal thrombocythemia not only occur in the extremities as asymmetric red (erythros), burning painful (algos) and congested toes finger (extremities = melos) but also involve the ocular, cerebral and coronary circulation with clinical manifestations of migraine-like atypical transient

ischemic attacks (MIAs) (Figure 4). In contrast, PE or inherited erythromelalgia spontaneously arises in early childhood or adolescence and persists throughout life. The red (erythros) burning pain (algos) of PE is typically initiated or aggravated by warmth and exposure to heat (therme) with locations in the skin of the body and legs with sparing of the toes, fingers (extremities). Consequently the term PE in Nav1.7 neuropathy is best replaced by inherited and incurable erythrothermalgia (ETM) derived from the words erythros = red, therme = warmth and algos = burning pain in the absence of microvascular pathology (Figure 5). As NAV1.7 protein is not expressed in central nervous system erythrothermalgia is not accompanied by mental disturbances. Nav1.7 protein is expressed in dorsal root ganglions (DRG), in nociceptive peripheral neurons [110–113], in smooth muscle cells of cutaneous arterioles, arteriole-venule (AV) shunts and in endothelial cells lining arterioles and AVS [64]. Consequently, the Nav1.7 gain of function mutation electrophysiological disturbances in ETM patients do occur not only in small fiber DRG neurons but also in smooth muscle cells of the arteriole-capillary-arteriole-venous-shunt functional unit and predicted to have excitability characteristics similar to nociceptors of DRG neurons [110–113]. Incurable inherited neuropathic Nav1.7 channelopathy has been recognized in this report by Michiels as a vasomotor disorder of the neurosensory-vasomotor AV shunt as part of the arteriolar-capillary-AVS function unit in the skin of the body, feet and toes (Figure 5) [63,64].

4.2. Molecular pathobiology of PE or ETM in neuropathic Nav1.7 channelopathy

Nav1.7 is an integral membrane protein that consists of four Domains (D1–D4), and each D domain has a S1–S4 voltage sensing structure. S4 has positive amino acids that act as gating charges for permeation pores for Na⁺ current. S4–D1D2 determines voltage dependence of activation, S4–D3D4 contributes to coupling activation and inactivation and S5 and S6 form the locus of ion selection. With the exception of a few, there are no ETM and PEPD mutations in the S1–S2–S3 of each D1 to D4 domains. ETM and SNF Nav1.7 mutations are mainly distributed over S4–S6 and Loop L1–L2 of D1 and D2 [70,111–113]. PEPD Nav1.7 mutation are mainly distributed over L2, S4–S5, inner' D3, L3 and D4. ETM Nav1.7 mutations in transfected HEK cells display hyperpolarized activation by increase of neuron excitability through: (i) hyperpolarized shift in inactivation, (ii) depolarized shift in steady state inactivation, (iii) slowed deactivation and (iv) RAMP currents (Table 5). Increased overlaps between activation and inactivation curves enhance RAMP current in Nav1.7 transfected HEK cells. Nav1.7 is expressed in the majority of small diameter C-fiber with unmyelinated slow conducting DRG neurons, and less in medium and large diameter A-fibers with rapid conducting heavy myelinated DRG neurons. Functional analysis using patch clamp recording showed that Nav1.7 mutations cause a hyperpolarizing shift in activation of the sodium channel and a slowing of deactivation [70,111–113]. This is accompanied by an enhanced response to small depolarizing changes that should confer hyperexcitability on small fiber neuron cells.

The Nav1.7 sodium channels are not globally present within all neurons but are selectively expressed within peripheral small C-fiber sensory neurons in dorsal root ganglia (DRG, Figure 5). The ETM linked SCN9A gene mutations studied to date lower the threshold for single action potentials and increase the firing frequencies (spontaneous action potentials = DRG-SAP in Table 6) of transfected DRGs with many ETM linked SCN9A mutations causing a depolarizing shift in resting potentials [111–113]. Nav1.7 mutations contribute to the hyperexcitability of pain-signaling DRG neurons thus causing extreme sensitivity to pain. Genotype–phenotype analysis of the SCN9A gene showed that smaller effects on sodium channel activation are related to a smaller degree of DRG neuron excitability and later onset of clinical ETM signs and symptoms [111–113].

4.3. Diagnostic differentiation of PEPD and ETM nav1.7 channelopathies

Sudden attacks of rectal, ocular, and submaxillary pain, first described in 1958, has been labeled by Fertleman et al. as paroxysmal extreme pain disorder (PEPD) [114,115]. PEPD is an autosomal dominant disorder characterized by paroxysmal episodes of sudden onset excruciating erythromelgic pain like fire in severe cases at different body sites, accompanied by skin flushing. The onset of PEPD is in the neonatal period or very early infancy, persists throughout life and autonomic symptoms predominate initially with skin flushes in all and harlequin color changes and tonic cramps in most. Syncope with bradycardia and sometimes asystole are common. Later, PEPD is featured by erythrothermalgic attacks of deep burning pain in the rectal, ocular or jaw areas, but also diffuse. Attacks are triggered by defecation, cold wind, eating and emotion. PEPD is frequently misdiagnosed as epilepsy, hyperplexia or reflex anoxic seizures in early childhood. PEPD spontaneously ameliorates at older ages in adolescence and in adults. A genome wide-linkage search in one large pedigree with PEPD led to the PEPD susceptibility gene on chromosome 2q24.3 [114]. As this region contained SCN9A, the investigators sequenced the gene and found eight heterozygous Nav1.7 sodium channel missense mutations in eight families (R996 C, V1298 F, V1299 C, I1461 T, I1462 V, T1464I, M1627 K) [115]. Each family possessed a unique PEPD mutation and one individual was compound heterozygous for R996 C and V1298 F. This individual was more severely affected than his heterozygous father. Functional analysis of three gain of function mutations (I1461 T, T1464I and M1627 K) were reported to impair inactivation of the Nav1.7 channel. There are clear mechanistic difference between PEPD and ETM [116,71]. PEPD and ETM have entirely distinct phenotypes as the result of distinct repertoires of mutations leading to either lowered threshold of activation in ETM or defective inactivation in PEPD. ETM primarily affects the extremities in which the main triggers are warmth (temperature changes), exercise, and closed shoes. ETM mutations cause a significant hyperpolarizing shift in activation, slow deactivation and increased response to ramp depolarization (Table 5). In PEPD the

activation threshold is mostly normal. PEPD mutations cause a depolarizing shift rightwards in steady state inactivation and fast inactivation is incomplete followed by resurgent current leading to the generation of persistent sodium current. Fast inactivation of Nav1.7 is mediated by obliteration of the channel pore with the inactivation particle formed by the IFM motif on L3 between D3 and D4. This knowledge explain why the A1632 T Nav1.7 mutant showing no resurgent current is associated with IEM at age 3 and 7, and why the *de novo* mutant A1632E showing resurgent currents is associated with neonatal PEPD and ETM at age 3 [101,102]. Nav1.7 mutations associated with PEPD, but not ETM, enhance Nav beta-4 peptide-mediated resurgent sodium currents [71]. The Nav1.7 mutations R185 H, G856D, M932 L/V991 L and M1532I presented with an overlap clinical phenotype of ETM and PEPD and did show the Nav1.7 beta4 peptide-mediated resurgent sodium currents [70,87,71]. All but a few ETM mutations are localized in S4-S5-S6 of the D1 and D2 domains. All but 1 (R996 C) PEPD mutations are localized in the S4-S5 linkers and L3 of the D3 and D4 domains. Fast inactivation of Nav1.7 is mediated channel pore obliteration with the inactivation particle formed by the IFM motif on L3 between D3 and D4, whereas parts of the S4-S5 segment serve as inactivation particle receptor sites. Most PEPD mutations can be found in areas involved in fast inactivation: L3 and the S4-S5 linkers [71]. The PEPD mutations very likely disturb the interaction of the IFM motif with the intracellular channel parts that act as its receptor, thereby impairing normal inactivation (receptor-function unit). Wild type Nav1.7 generates no or very small resurgent DRG current. DRG neurons transfected with PEPD mutant Nav1.7 yield robust resurgent current, indicating a contribution to hyperexcitability of DRG neurons. A PEPD mutation may render Nav1.7 prone to produce resurgent current, as it slows current decay [71]. Theile and Cummins examined whether carbamazepin versus the sodium channel inhibitors riluzol and anandamid exhibit differential inhibition of Nav1.7beta4 mediated resurgent currents in the PEPD mutants T1464I and M1627I [117]. Carbamazepin did not selectively target resurgent currents, whereas anandamid strongly inhibited resurgent currents with minimal effect on the peak transient current amplitude, thereby demonstrating that resurgent current can be selectively targeted [117]. These data provide evidence for a different role of the N-terminal and the C-terminal of the Nav1.7 sodium channel protein of the C-fiber afferent nociceptive DRG neurons. All PEPD mutations impair fast inactivation and show resurgent current, whereas the majority of ETM mutations located in S4-S5 of D1 and D2 decreased the threshold of activation and lack resurgent current [110, 111, 112, 113,71].

The electrophysiological alterations in Nav1.7 mutated SNF patients are more subtle with impaired slow inactivation as the most common change in Nav1.7 gating [116]. In three SNF Nav1.7 mutations (I739 V, I720 K and I228 M) impaired slow inactivation is the only detected change and one displays an additional small shift of steady state inactivation (D623 N) [70,71] (Table 5). Expression of SFN-Nav1.7

mutations in DRG neurons led to hyperexcitability indicating that impaired slow inactivation is enough to cause the erythrothermalgia (ETM) Nav1.7 channelopathy by itself.

4.4. Congenital insensitivity to pain (CIP) caused by loss of function mutations in the SCN9A gene

Cox et al. discovered SCN9A loss of function mutations that cause complete loss of Nav1.7 excitability of DRG neurons in six patients from three consanguineous Japanese families with a congenital insensitivity to pain (CIP) [118]. The null mutations were identified in exon 10 (S495X), exon 13 (I767X), and exon 15 (W897X) of the SCN9A gene [118]. A second study in nine Western European and North and South American families [119] and a third large Canadian family [120] used linkage analysis to search for homozygous haplotypes identified the same gene and detected 10 truncating SCN9A mutations. The majority of affected patients in these two studies were homozygous for SCN9A mutations and two patients were compound heterozygous for different SCN9A nonsense mutations. All CIP patients had injuries by biting themselves in their lips frequently requiring plastic surgery and in their tongue with loss of the distal third during the first few years of life frequent bruises and most had suffered traumatic fractures complicated by osteomyelitis. CIP children were considered of normal intelligence by their parents, teachers and physicians. Neurological examinations reveal normal sensations of touch, warm and cold temperature, proprioception, tickle and pressure, but not for painful stimuli of burning and trauma. Body strength, muscular tone, reflexes (plantar responses, corneal and gag reflexes) and joints are normal. All CIP patients had normal health, vision, hearing and appearance and normal autonomic nervous system functions.

Nonsense (null) mutations in Nav1.7 that cause CIP are randomly distributed over D1, D2, D3 and D4 domains (R277X, Y328X, E693X, I767X, R830X, W897X, F1200 L, I235LfsX2, C.4336-7-10delGTTT, R1488X, M1488X, M1659X, W1689X) [110,111,118, 119, 120]. Affected CIP individuals are either homozygous or compound heterozygous for SCN9A (nonsense = null) mutations with the complete absence of Nav1.7 sodium channel membrane protein. Affected CIP individuals with the complete absence of Nav1.7 sodium channel membrane protein are unable to feel physical pain from wounds or burns, have intact ability to distinguish between sharp and dull stimuli and to detect differences in temperature. CIP individuals are insensitive to pain beginning in infancy that leads to frequent painless injuries, fractures, burns, ultimately resulting in a shortened life span. CIP patients have intact sensory modalities, motor, cognitive, smell and sympathetic nerve functions, normal nerve conduction, tactile sense and vibration sense. Yuan et al. reported three novel missense mutations (M899I, M932 L and V1104 L) in two children with recessive CIP [121]. For case 1 a new heterozygous M899I mutation was identified. For case 2 two novel double heterozygous mutations (M932 L/V1104 L) were found. The two CIP patients were not completely painless but did have little pain from injuries. M899I and M932 L were not found in the family members, whereas V1104 L was found in 6.5% of control subjects [121].

The V1104 L mutation may be associated with diminished nociceptor excitability and lower pain perception in the general population mutation [121]. Drenth et al. (2005) described a sporadic case of primary erythermalgia (PE) with the R1150 W mutation, which was not found in 100 controls [79]. Harty et al. described the R1150 W in an affected proband carrying the A863 R gain of function mutation [90]. The R1150 W was presenting several unaffected family members (father and brother of proband A863 R) and in 14% of ethnically matched controls. Interestingly, the R1150 W mutation doubled the firing frequency in transfected DRG neurons [122] similar as in Q10 R Nav1.7 mutation in a patients with ETM with onset of pain at 14 years [65]. These observations suggest the possibility of disease-modifying genes or some other SFN defect that increase the penetrance of PE or ETM to 100% in the case of R1150 W [113,65,122]. Reimann et al. demonstrated that R1150 W quantitatively alters pain sensitivity and pain experienced in common conditions such as osteoarthritis and spinal root pain [123]. Harrer et al. described a family with autosomal dominant R1150 W mutation in the SCN9A gene and Nav1.7 protein with a distinct late onset (46 years) primary erythermalgia (PE relabeled as ETM) with regard to the pain description in whom the erythrothermalgia was elicited by exercise and warmth without skin changes, redness and swelling of the leg [124].

4.5. Quantitative skin testing (QST) and biopsy for the diagnoses of ETM and SNF anno 2016

The clinical phenotypes and natural histories of patients with Nav1.7 mutated incurable ETM, PEPD [114,115] and SFN-neuropathy [70,71] are well known. The positive predictive value quantitative skin testing (QST) for reduced intradermal nerve fiber density in skin biopsies as a gold standard for diagnosis of SFN is high [125,126]. McDonnell et al. reported on QST profiles of affected and unaffected sites of 13 ETM patients with the mutation S241 T, F1449 V and V400 M. Subjects described their pain as ‘burning’, like being on fire’, constant pain, like frostbite that is warming up, blowtorch-like, hot, throbbing, aching, tingling or pins and needles. Exercise and heat were the most common triggers for pain followed by hot humid weather, physical activities and wearing closed shoes or socks. In our experience cooling the feet and legs in ice-cold water worsened the painful condition complicated by skin lesions and infection. There was no consistent relationship between the number of pain attacks per day and severity of pain between attacks for each subject in the study of McDonnell et al. [126]. Scores for cold pain threshold (CDT), warm detection threshold (WDT) and thermal sensory limen (TSL) were lower in skin areas of affected sites as compared to unaffected sites suggesting reduced (impaired) small fiber function in affected skin areas in all three neuropathic ETM mutants studied. During attacks cold pain threshold (CPT) and heat pain threshold (HPT) decreased from above to below zero-score. Regarding thermal pain thresholds, scores for cold and heat threshold (CPT, HPT) were significantly higher at the unaffected site compared to the affected site indicating defective small fiber neuropathy.

This points to the need to perform functional, histopathological and histochemical studies of the neuron-arteriole-capillary AVS function unit in biopsies from affected and unaffected skin areas to address the question why hyperalgesia to cold and heat stimuli is more pronounced in unaffected as compared to affected skin areas. Mechanical pain threshold (MPT) and mechanical pain sensitivity (MPS) were and remained normal in affected and nonaffected and in affected sites outside and during attacks in the three neuropathic ETM channelopathies studied. Skin biopsies for the evaluation of small fiber neuropathy has never been performed in ETM patients including the eight S241 T, V400 M and F1446 ETM patients in the study of McDonnell et al. [126]. A systematic evaluation of QST, skin biopsy, related to lowering of activation threshold, depolarized slow and fast inactivation and impaired deactivation in Nav1.7 transfected HEK cells and resurgent currents and firing in Nav1.7 transfected DRG as well as real life studies on the neurovasomotor disturbances of the arteriole-capillary-AVS functional unit in the skin is warranted for all neuropathic ETM and PEPD patients. Neuroscientists, pharmacologists, biomedical technologists, neurophysiologists, geneticists and molecular biologists in partnership with multidisciplinary internists, neurologists, dermatologists and cardiologists are obliged to collaborate in scientific friendship to study affected patients of all variants of sodium channelopathies Nav1.1 to Nav1.9 in great detail trying to make significant progress for novel Nav-channelopathy-inhibitor treatment options to save lives of Bruguda patients and solve unbearable pain of affected neuropathic ETM and PEPD patients [127,128].

5. Expert opinion: common etiologic pathways in aspirin responsive erythromelalgia and incurable erythrothermalgia

5.1. Incurable inherited neuropathic erythrothermalgia: ETM

Nav1.7 protein is expressed in dorsal root ganglions (DRG), in nociceptive peripheral neurons [110, 111, 112, 113], in smooth muscle cells of cutaneous arterioles, arteriole-venule shunts (AVS) and in endothelial cells lining arterioles and AVS (Figure 5) [64]. The gain of function mutations in the SCN9A gene in PE/ETM and PEPD patients result in electrophysiological disturbances of the Nav1.7 mutated small fiber DRG nociceptive neurons and the smooth muscle cells of the arteriole-capillary-arteriole-venous-shunt (AVS) functional unit (Figure 5) [110, 111, 112, 113]. Nav1.7 channelopathy has been recognized in the report as a painful vasomotor disorder of the neurosensory-vasomotor arteriolar-capillary-AVS functional unit in the skin of the body, feet and toes (Figure 5) [63,64]. As Nav1.7 protein is expressed in smooth muscle cells and the afferent C-fibers of the cutaneous arterioles and arteriole-venule shunt, the gain of function mutation in the Nav1.7 channelopathies makes the smooth muscle cells and its afferent innervation by C-fibers in the arteriole-capillary-AVS hyperexcitable in patients with PE, ETM and PEPD [64]. Sensory DRG neurons in PE/ETM and PEPD patients display

hyperexcitability DRG neurons showing spontaneous firing of action potentials as the cause of incurable inherited neuropathic PE, ETM and PEPD erythromelalgia variants. The burning pain like fire in severe cases of incurable neuropathic erythrothermalgia in Nav1.7 channelopathy [29,9], can readily be labeled as the 'Man on Fire Syndrome' (Figure 5). The facts that Nav1.7 sodium channel has low expression in the brain can readily explain the lack of epileptic seizures in PE, ETM and PEPD patients. Fortunately PE, ETM and PEPD patients have normal senses modalities of large myelinated nerve fibers, which are not part of the nociceptive (pain-signaling) dorsal root ganglia (DRG) nociceptive neurons of the vasomotor-neurosensory arteriolar-capillary-AVS functional unit in the skin all over the body, face, leg and feet [64].

5.2. Aspirin responsive erythromelalgia in thrombocythemia

The initial stages of platelet-mediated aspirin erythromelalgia in clonal thrombocythemia are featured by livedo reticularis of the skin, acroparesthesias of prickling 'needles' sensations and caused by shear stress induced activation of hypersensitive JAK2 mutated platelets in the endarterial microcirculation (Figures 4 and Figure 6) followed by platelet mediated arteriolar occlusive thrombosis if left untreated (Figure 2). The release of prostaglandin endoperoxides, serotonin, and adenosine accounts for the inflammatory signs, burning pain and red congestion of typical erythromelalgia (Figure 4). In this ongoing process of platelet activation, platelet aggregation and platelet-endothelial interactions, the endothelial cells release thrombomodulin indicating endothelial cell activation and damage. Release of PDGF by activated platelets account for the fibromuscular intimal proliferation (Figures 2 and 4). The analgetic effect of aspirin immediately and completely relieves the inflammatory, burning painful vasomotor manifestations for a few days by irreversible inhibition of platelet cyclooxygenase (COX-1) [20,42] and subsequently will prevent the erythromelalgic microvascular complications through maintained irreversible inhibition of platelet cyclooxygenase (COX-1) by low dose aspirin 50–75 mg daily [37]. The analgesic agents sodium salicylate and paracetamol and anticoagulation with coumadin and heparin do not inhibit platelet cyclooxygenase and are ineffective to relieve the pain and inflammatory manifestations of erythromelalgia [20,37,42]. The high shear rate of blood flow in arterioles as compared to that in arteriolar circulation contributes to the location of erythromelalgia and MIAs in the arteriolar endarterial circulation in thrombocythemia [20,42] caused by a gain of function mutation in the JAK2, MPL and TPO genes [103,104,46, 47, 48, 49, 50, 51, 52, 53, 54, 55]. Spontaneous activation and reversible aggregation of hypersensitive JAK2 mutated thrombocytic platelets in the arteriolar circulation is a pathophysiological process which transforms platelet membrane phospholipid by phospholipase A2 into arachidonic acid as the source for cyclooxygenases (COX-1) to produce prostaglandin endoperoxides and the release by platelet of serotonin, platelet factor 4, beta-thromboglobulin, PDGF, adenosine diphosphonate. The inflammatory mediators, such as prostaglandin [60], adenosine [61] and serotonin [62] directly affect

the afferent (pain-signaling) nociceptive C-fiber neurons of the vasomotor-neurosensory arteriolar-capillary-AVS functional unit in the skin and induce erythromelalgic pain and inflammatory manifestations of red warm and congested skin (Figures 2 and 4), which can only be reversed by irreversibly inhibition of platelet COX-1 [38] with aspirin as the explanation of relief of erythromelalgic pain and inflammatory manifestation [20,42] in JAK2, MPL and TPO mutated thrombocythemia ('one molecular cause etiology of aspirin responsive erythromelalgia'). Other platelet inhibiting drugs like sulfinpyrazon, dipyridamol, ticlopidine, and clopidogrel do not inhibit platelet COX-1 activity and consequently have no effect on erythromelalgia in thrombocythemia [20,42].

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Declaration of interest

The author is a founder of the Goodheart Institute & Foundation in Nature Medicine & Health, Rotterdam, The Netherlands, Freedom of Science and Education, European Free University. JJ Michiels serves as consultant professor in the Bloodcoagulation, Hemostasis Research Laboratory (co-founder VWF-VWD research program) at the department of Hematology University Hospital, Antwerp; as consultant professor in Hematology and Bloodcoagulation, Comenius University, Bratislava and Martin, Slovakia; as consultant professor in Hematology, Romania MPN Working Group, University Hospital Bucharest; consultant to the Dutch Society of Internal Medicine and Ministry of Public Health; consultant of Industrial and Pharmaceutical Medicine; as an editor of 3 Medical Journals and as a guest editor on request and by self initiation. Writing assistance was utilised in the preparation of this manuscript, it was carried out at the Goodheart Institute and Foundation in Nature Medicine and Health, Rotterdam, The Netherlands. The author has no other relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript apart from those disclosed.

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